

ENVS193DS-Final

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Repository Link

[GitHub Repository](#)

Set Up

Packages:

```
# reading in packages
library(tidyverse)
library(here)
library(janitor)
library(readxl)
library(here)
library(scales)
library(ggeffects)
library(MuMIn)
library(forcats)
library(DHARMA)
library(ggeffects)
```

Data:

```
# creating object from 'nests' from data
nests <- read.csv(here('data', 'nest_data_final.csv'))
```

Problem 1

a.

In part 1 of the results section, the co-worker used a linear regression. In part 2 of the results section, the co-worker used a Chi-Squared test.

b.

Statement 1: Based collected American Avocet micro-habitat use and distance to water edge, American Avocet micro-habitat use is significantly influenced by proximity to water edge.

Statement 2: Based on collected shorebird species presence across the habitats managed pond, non-tidal marsh, salt pond, tidal marsh, and tidal mudflap, shorebird species significantly differ by locale.

c.

One additional variable that could influence the probability of American Avocet microhabitat use is the water depth surrounding habitats, a continuous variable measured in meters, collected via bathymetry or hands-on-measurement depending on depth. This variable might influence American Avocet habitat use as water depth is a significant driver of bird use in managed wetlands – water depth may influence American Avocet use depending on feeding patterns (deeper areas are inaccessible to wading birds), present food type (different depths may lead to different flora and fauna presence), and predator presence (predators may reside in varying depths of water/predators may not be able to access certain water depths), playing a role in determining how American Avocets use these locales (Schacter et. al., 11).

Problem 2

a.

```
# create new object called 'nests_clean' from 'nests'
nests_clean <- nests |>
  # clean names
  clean_names() |>
  # filter ant species
  filter(
    ant_species %in% c("RRB", "AB", "BBR")) |>
  # full names
  mutate(
    species_name_full = recode_values(ant_species,
      "RRB" ~ "Crematogaster mimosae",
      "AB" ~ "Crematogaster sjostedti",
      "BBR" ~ "Crematogaster nigriceps")) |>
  # reordering species
  mutate(
```

```

# order of species
species_name_full = fct_relevel(species_name_full,
                                "Crematogaster mimosae",
                                "Crematogaster sjostedti",
                                "Crematogaster nigriceps")) |>

# arrange in order
arrange(species_name_full) |>
# columns of interest
select(
  height_cm,
  case_control,
  species_name_full)

# display structure
str(nests_clean)

```

```

'data.frame': 270 obs. of 3 variables:
 $ height_cm      : num  392 310 513 154 413 ...
 $ case_control   : int   1 0 0 0 1 0 0 0 0 0 ...
 $ species_name_full: Factor w/ 3 levels "Crematogaster mimosae",...: 1 1 1 1 1 1 1 1 1 1 ...

```

```

# display 10 random rows
nests_clean |> slice_sample(n = 10)

```

	height_cm	case_control	species_name_full
1	236.5	0	Crematogaster mimosae
2	131.0	0	Crematogaster mimosae
3	286.0	0	Crematogaster sjostedti
4	199.0	0	Crematogaster mimosae
5	85.0	0	Crematogaster sjostedti
6	248.5	1	Crematogaster mimosae
7	213.0	0	Crematogaster mimosae
8	153.5	0	Crematogaster mimosae
9	410.0	0	Crematogaster mimosae
10	150.0	0	Crematogaster nigriceps

b.

In this data set, 1s and 0s indicate if a tree has a nest (1) or is “available” (0); in other words, they indicate if a tree contains a birds nest or not.

c.

Tree Height: Tree height is a continuous variable. According to paragraph two in the results subsection in **Symbiotic acacia ants drive nesting behavior by birds in an African savanna**, as tree height increases, probability of nest occurrence probability will increase as well, as higher nests decrease ground predator nest predation rates, increasing overall bird lifetime fitness.

Acacia Ant Species: Acacia ant species is a categorical variable (C.nigriceps, C.mimosae, and C.sjostedti). According to paragraph one in the discussion subsection in **Symbiotic acacia ants drive nesting behavior by birds in an African savanna**, the presence of more aggressive defending symbionts (C.nigriceps and C.mimosae) will result in an increased probability of nest occurrence probability, aggressive defender presence may reduce nest predation rates, and increase overall bird lifetime fitness.

d.

Model number	Ant Species	Tree Height	Canopy Area
1 (no term)	x	x	
2 (with term)	x	x	x
3 (species)	x		
4 (height)		x	
5 (canopy)			x

e.

```
# model 1
model_1 <- lm(
  case_control ~ height_cm + species_name_full,
  data = nests_clean_it)

# model 2
model_2 <- lm(
  case_control ~ height_cm + species_name_full + alive,
  data = nests_clean_it)

# model 3
model_3 <- lm(
  case_control ~ species_name_full,
```

```

        data = nests_clean_it)

# model 4
model_4 <- lm(
  case_control ~ height_cm,
  data = nests_clean_it)

# model 5
model_5 <- lm(
  case_control ~ alive,
  data = nests_clean_it)

```

f.

```

# running AICc
AICc(
  model_1,
  model_2,
  model_3,
  model_4,
  model_5
) |>
arrange(AICc)

```

The best model as determined by Akaike's Information Criterion (AIC) was the model using predictor variables ant species, tree height, and canopy area for the response variable of bird nest presence (model 2).

g.

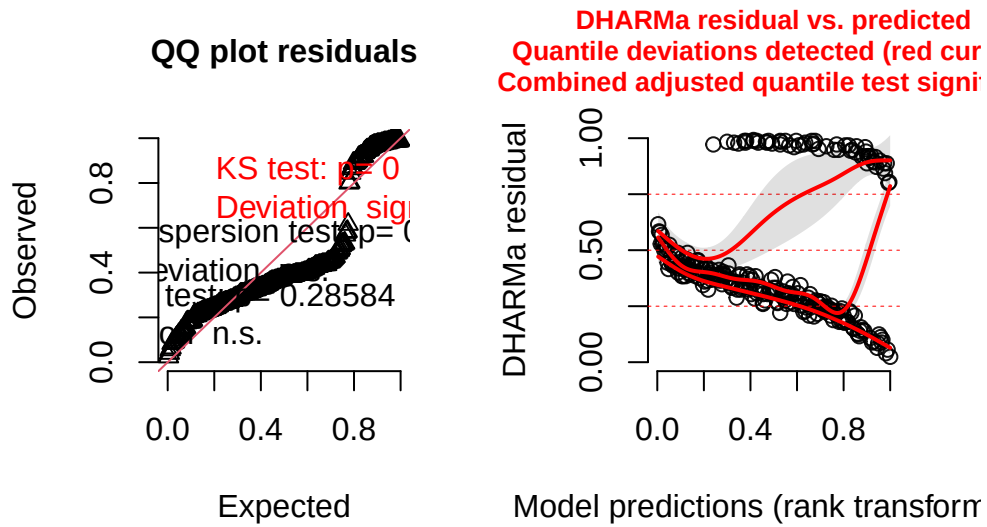
```

# running dharma diagnostic
diagnostic_checking <- simulateResiduals(fittedModel = model_1)

# display dharma diagnostic
plot(diagnostic_checking)

```

DHARMA residual



h.

```
# generating predictions
preds <- ggpredict(
  model_1,
  terms = c("height_cm", "species_name_full")) |>
# renaming columns
rename(height = x,
  species_name_full = group)
```

i.

j.

k.

Problem 3

a.

My visualizations use different metrics and different model types across Homework 2/3 and my affective visualization; in Homework 2 and 3 I used box plots to visualize my data, while I used a trendline to visualize my data in my affective visualization. Across my visualizations, I did not see many similarities – my affective visualization reflected my predicted outcomes with different variables, while my visualizations in homework 2/3 did not show the trends that I depicted in my affective visualization; similarly, I did not see many patterns across my affective and effective visualizations, likely due to different metrics of creation and data alignment. I did not receive any feedback on my Homework 2 and 3 visualizations, however, in workshop, I got a lot of visual suggestions – for instance, implementing a clearer trend line in my drawing – which was reflected in my final affective visualization (I added a red trend line, and used colors to create clearer visual distinctions between my data).

b.

Using the response variable of productivity % and the predictor variable of activity time (as expressed in my affective visualization as an aggregation of outside activity and class time), one analysis that we talked about this quarter that I could apply to my research question is a correlation analysis. Both of my variables are numerical continuous variables, with one variable (activity time) predicted to have an influence on the other variable (productivity %). Subsequently, I believe that a correlation analysis is appropriate in trying to answer the question: how do my activities impact my productivity?